Table 3. Conglomerate tests on the six boulder and diamicton localities (Fig. 7o–t). Specimen HT (ChRM) directions with MAD ≤ 15° were averaged to their parent boulder, block or core, and the unit means were tested for randomness with the Watson (1956) resultant-length test. Ro95 and Ro99 are the critical resultant lengths for N random unit vectors from 2×10^5 Monte-Carlo draws. R < Ro95: the clast directions are random and the clasts were magnetized before deposition (test passes). R ≥ Ro95: they are clustered (test fails). Clast mean = Fisher mean of the unit means. The lines-and-planes column repeats the test with unresolved specimens treated as remagnetization circles (McFadden & McElhinny, 1988). PKQ-A is tested on its three field boulders (cores 0–4, 5–6 and 7–9). Tests with N = 3 (NWF-B, PKQ-A, PKS-H) have little power. Shaded rows fail.

Locality	Units tested	N (n spec)	R	Ro95	Ro99	Clast mean D/I (°)	k	α95 (°)	R (lines + planes)	Verdict
NWF-B	Blocks (3)	3 (20)	2.38	2.62	2.83	76/−60	3.2	85	2.38	Random
NWF-C	Boulders (8)	8 (33)	3.26	4.47	5.24	3/+22	1.5	77	3.26	Random
PKQ-A	Field boulders (3; 10 cores)	3 (82)	2.25	2.62	2.83	146/+48	2.7	–	2.25	Random
PKS-F	Boulders (11)	11 (105)	5.31	5.28	6.25	36/−15	1.8	51	6.09	Clustered
PKS-G	Boulders (8)	8 (33)	6.58	4.47	5.24	159/+4	4.9	28	6.58	Clustered
PKS-H	Cores HA1, HA2, HB1	3 (24)	2.52	2.62	2.83	184/−4	4.1	71	2.50	Random

Table 4. Posterior mixing weights (wt%, mean ± 1σ combining within- and between-chain variance) of each lithology type in the reversible-jump MCMC unmixing of the 82 accepted EPMA analyses (Fig. 5k, m). The number in parentheses is the percentage of posterior samples in which the type is present. Full model: 12 chains × 6 × 10^4 iterations; sensitivity runs: 4 chains × 3 × 10^4 iterations. "Excluded" marks types removed from the dictionary in that run.

Lithology type	Full model (12 chains)	No random (4)	Na_2O, K_2O excluded (4)	Regional only (4)
Khorat sandstone	35.6 ± 15.4 (91)	26.2 ± 13.3 (99)	40.1 ± 11.9 (100)	25.0 ± 11.2 (99)
Khorat mudstone	3.3 ± 5.6 (37)	32.2 ± 20.8 (100)	3.8 ± 4.6 (48)	25.5 ± 12.0 (100)
Unweathered shale (PAAS)	26.8 ± 23.1 (91)	7.2 ± 8.5 (50)	28.2 ± 15.8 (100)	excluded
Felsic metatuff	9.9 ± 5.7 (91)	14.7 ± 11.1 (75)	6.2 ± 6.3 (65)	30.6 ± 12.9 (100)
Bolaven tholeiite	3.7 ± 5.2 (53)	10.2 ± 5.1 (100)	7.3 ± 9.5 (50)	6.3 ± 5.5 (71)
Bolaven alkali basalt	1.1 ± 2.7 (19)	8.6 ± 9.7 (75)	0.0 ± 0.2 (1)	11.9 ± 6.3 (86)
All basalt (sum)	4.8 ± 5.4 (64)	18.8 ± 9.4 (100)	7.3 ± 9.5 (50)	18.2 ± 2.3 (100)
Lateritic saprolite	0.0 ± 0.0 (0)	0.0 ± 0.0 (0)	0.0 ± 0.0 (0)	0.0 ± 0.0 (0)
Sedimentary laterite / bauxite	1.6 ± 3.8 (17)	0.0 ± 0.0 (0)	0.0 ± 0.0 (0)	0.0 ± 0.0 (0)
Diamicton / loess	0.4 ± 1.6 (6)	1.0 ± 3.0 (14)	2.4 ± 3.5 (40)	0.7 ± 2.1 (12)
Random (unconstrained)	17.8 ± 13.2 (86)	excluded	12.0 ± 8.7 (75)	excluded

Table S1. EDS analyses of Bolaven basalt specimen LS-H18 — see the existing table (formerly Table 3).

Table S2. Hysteresis parameters of the seven fresh-interior / weathered-rind chip pairs plotted in Fig. 4c (MicroMag VSM; Hc and Mr/Ms from the slope-corrected loops, Hcr from backfield remanence). Shaded rows are weathered chips. PKS24-I and PKS24-HB were filed as PKQ24-… in the instrument files.

Site	Setting	Sample	Chip	Mass (g)	Hc (mT)	Hcr (mT)	Mr (μA m²)	Ms (μA m²)	Mr/Ms	Hcr/Hc
PKS-B	In-situ flow	PKS24-BA-P1	fresh	0.299	16.8	24.5	33.6	96.7	0.348	1.46
			weathered	0.164	17.3	34.2	8.6	32.5	0.265	1.98
PKS-H (HB)	In-situ flow	PKS24-HB-P1	fresh	0.423	14.7	23.3	30.7	97.5	0.315	1.58
			weathered	0.023	10.7	18.7	2.1	8.3	0.258	1.75
PKS-I	Corestone	PKS24-I-P1	fresh	1.307	26.2	47.5	90.4	304.5	0.297	1.81
			weathered	0.108	22.2	42.1	23.3	86.5	0.270	1.90
NWF-C	Diamicton clast	NWF24-C-P8	fresh	0.307	21.1	38.3	25.5	78.0	0.327	1.82
			weathered	0.300	16.6	32.6	28.8	108.3	0.266	1.97
PKQ-A (2A)	Diamicton clast	PKQ24-A-P2A	fresh	0.637	15.7	33.0	34.2	127.4	0.268	2.10
			weathered	0.359	10.6	19.9	32.2	173.4	0.186	1.88
PKQ-A (5A)	Diamicton clast	PKQ24-A-P5A	fresh	0.790	13.9	30.3	34.1	132.0	0.258	2.17
			weathered	0.286	8.4	16.8	19.2	120.0	0.160	1.99
PKQ-A (5C)	Diamicton clast	PKQ24-A-P5C	fresh	0.668	10.6	23.2	27.4	133.0	0.206	2.18
			weathered	0.245	8.7	17.1	15.6	90.8	0.171	1.97

Table S3. Paleomagnetic sample inventory. Coordinates are means of the 2024 GPS waypoints (WGS84). Orientation from the sample-header files: sun compass, or magnetic compass corrected to IGRF declination. Specimens: thermally demagnetized specimens with fitted components; HT accepted: specimens whose HT (ChRM) fit has MAD ≤ 15°; component fits: all LT, MT and HT fits.

Site	Lat (°N)	Lon (°E)	Setting	Sample type	Orientation	Cores / blocks	Specimens	HT accepted (MAD ≤ 15°)	Component fits
NWF-A	15.6446	106.1427	In-situ flow	Block	Sun	5	15	15	40
NWF-B	15.6236	105.9859	Blocks (transported)	Block	Magnetic (IGRF-corr.)	3	20	20	58
NWF-C	15.4148	106.1433	Diamicton boulders	Block	Sun	8	33	33	97
PKQ-A	15.1020	105.8536	Diamicton boulders	Core	Sun (cores 0, 3 magnetic)	10	83	82	238

Site	Lat (°N)	Lon (°E)	Setting	Sample type	Orientation	Cores / blocks	Specimens	HT accepted (MAD ≤ 15°)	Component fits
PKS-B	15.1543	106.1897	In-situ flow	Block	Sun	1	7	7	21
PKS-F	15.1131	106.2282	Diamicton boulders	Core	Sun	11	112	105	334
PKS-G	15.1558	106.1903	Diamicton boulders	Block	Magnetic (IGRF-corr.)	8	33	33	98
PKS-H	15.1557	106.1918	Two horizons (HA, HB)	Core	Sun	3	24	24	68
PKS-I	15.2328	106.1908	Corestone	Core	Sun	1	5	5	15
PKS-J	15.2431	106.2581	In-situ flow	Core	Magnetic (IGRF-corr.)	3	12	12	36
PKS-K	15.2419	106.2585	In-situ flow	Core	Magnetic (IGRF-corr.)	2	9	9	26
WBE-A	15.2700	106.3472	Basal flow	Block	Magnetic (IGRF-corr.)	2	11	11	31

Table S4. Tektites analysed in this study. EPMA spots with oxide totals below 97 wt% are rejected (shaded: no accepted analyses). LA-ICP-MS: five ablation sites with five replicates each.

Tektite	Locality	EPMA spots	Accepted (≥ 97 wt %)	Total range (wt%)	LA-ICP-MS
KK24_T1	Khon Kaen, NE Thailand	30	30	97.7–99.7	—
MG24_B1	Muong Nong area, Savannakhet, Laos	6	5	96.8–98.6	—
MNG24_A3_1	Muong Nong area, Savannakhet, Laos	30	30	98.4–99.7	yes (5 sites × 5)
MNG24_A3_2	Muong Nong area, Savannakhet, Laos	27	0	32.7–85.6	yes (5 sites × 5)
MNG24_A3_3	Muong Nong area, Savannakhet, Laos	15	0	60.7–96.7	—
MNG24_B1	Muong Nong area, Savannakhet, Laos	20	17	96.3–98.0	—
MNG24_C2	Muong Nong area, Savannakhet, Laos	25	0	65.8–93.9	—